UNIT-1

- **Kepler's Laws of Planetary Motion**

Kepler's law of planetary motion is the basic law that is used to define the motion of planets around the stars. These laws work in parallel with Newton's Law and Gravitation Law and are helpful in studying the motion of various planetary objects.

- Kepler’s laws of planetary motion were proposed by Johannes Kepler.
- These laws describe how planets move around the Sun in their orbits.
- The first law states that planets move in elliptical orbits with the Sun at one focus.
- The second law states that a line joining a planet and the Sun sweeps out equal areas in equal intervals of time.
- The third law states that the square of the orbital period of a planet is proportional to the cube of the semi-major axis of its orbit.

Kepeler's law provides three basic laws which are,

- Kepler's First law (Law of Orbits)
- Kepler’s Second law (Law of Areas)
- Kepler’s Third law (Law of Periods)

- **Introduction to Kepler's Law**
- The motion of an object is always relative with respect to other motions. Motions are divided into two categories based on the energy of the particle in motion:
- Bounded Motion
- Unbounded Motion

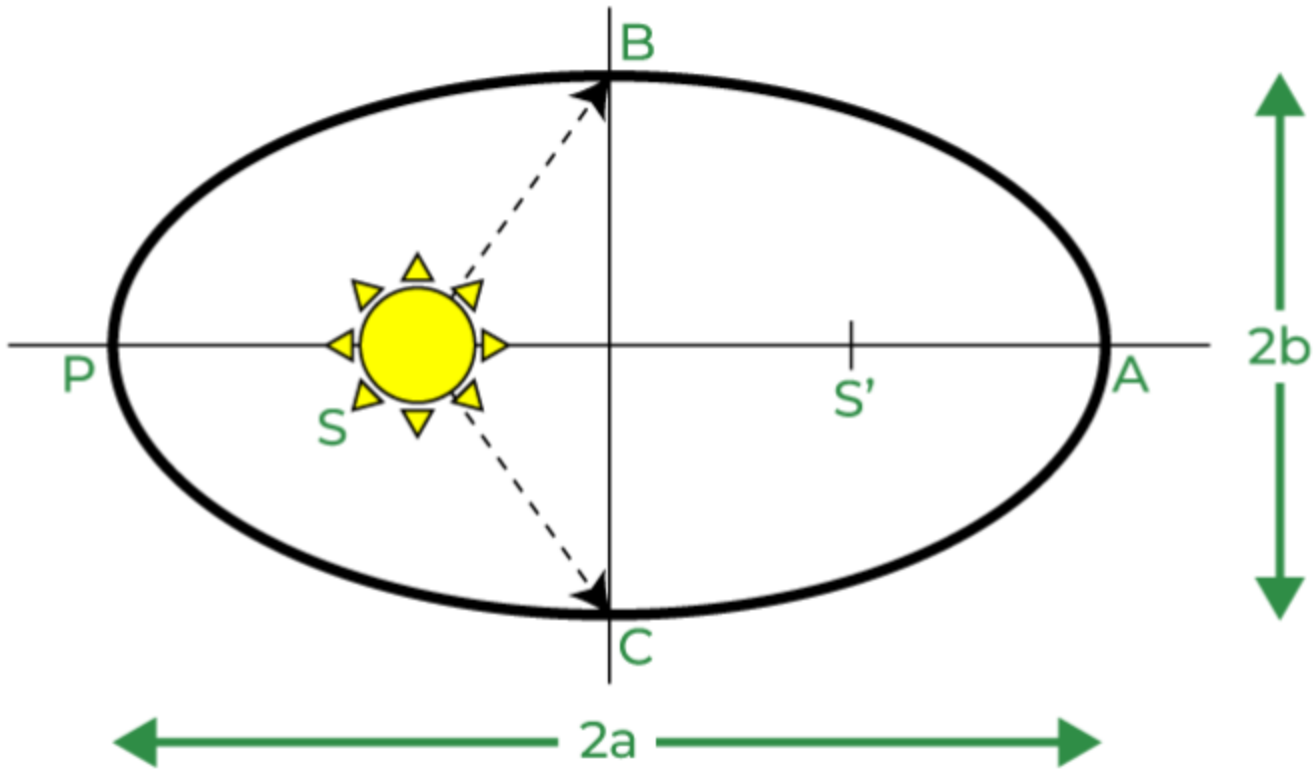
Bounded Motion

- The particle possesses negative total energy ($E < 0$) in bounded motion and has two or more extreme points where the total energy is always equal to the particle's potential energy, i.e. the particle's kinetic energy becomes zero.
- When the eccentricity $0 \leq e < 1$, $E < 0$ denotes that the body's motion is bounded. The eccentricity of a circular orbit is zero, whereas the eccentricity of an elliptical orbit is one.

Unbounded Motion

- The particle in unbounded motion has positive total energy ($E > 0$) and a single extreme point where the total energy is always equal to the particle's potential energy, i.e. the particle's kinetic energy becomes zero.
- When the eccentricity $e \geq 1$, $E > 0$ denotes that the body is moving freely. The eccentricity of a parabolic orbit $e = 1$, while the eccentricity of a hyperbolic orbit $e > 1$.
- Law of orbits, the law of equal areas, and the law of periods are Kepler's first, second, and third laws described in detail below.

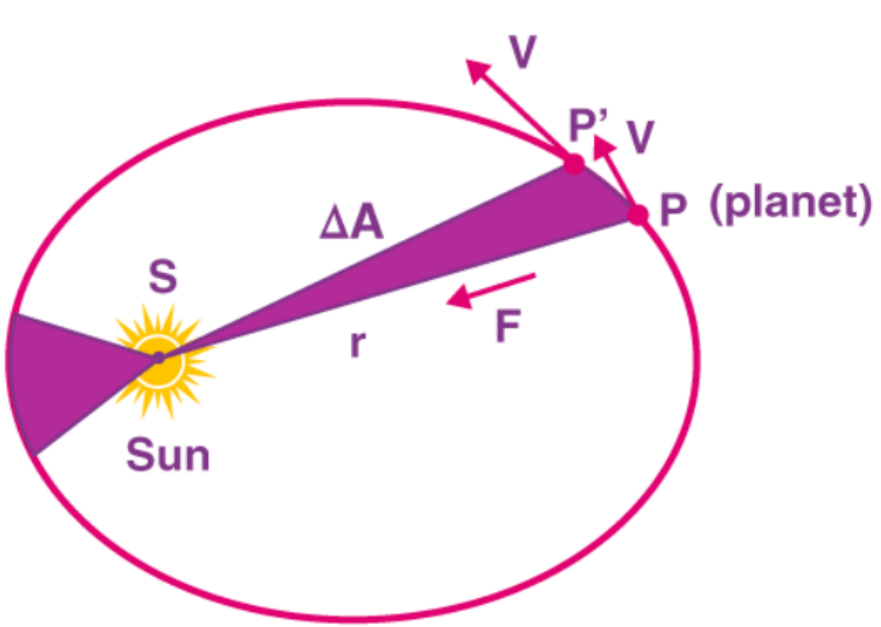
- **Kepler's First Law: The Law of Orbits**
- Kepler's First Law known as the law of ellipses, states that planets revolve around the sun in an elliptical pattern. The law of orbits is a common name for Kepler's first law.
- *According to the rule, any planet's orbit is an ellipse around the Sun, with the Sun at one of the ellipse's two focal points.*
- Planets circle the Sun in a circular orbit, which we are already familiar with. Planets do, however, rotate around the Sun, although not in a circular orbit, according to Kepler. However, it is centered on an ellipse.



- There are two foci (S & S') in an ellipse. The sun (S) is located at one of the ellipse's focal points. Perihelion (PS) refers to the point when the planet is closest to the sun, whereas aphelion (AS) refers to the point where the planet is furthest from the sun. The major axis is (2a) and the minor axis is (2b). The total of any planet's distances from two foci is constant, which is one of the properties of an ellipse. Seasons are brought about by a planet's elliptical orbit.

Kepler's Second Law – The Law of Equal Areas

- Kepler's second law states, "The radius vector drawn from the sun to the planet sweeps out equal areas in equal intervals of time".
- As the orbit is not circular, the planet's kinetic energy is not constant in its path. It has more kinetic energy near the perihelion, and less kinetic energy near the aphelion implies more speed at the perihelion and less speed (v_{min}) at the aphelion. If r is the distance of the planet from the sun at perihelion (r_{min}) and at aphelion (r_{max}), then,
- $r_{min} + r_{max} = 2a \times$ (length of the major axis of an ellipse) . . . . . (1)



Kepler's Second Law – The Law of Equal Areas

Using the law of conservation of angular momentum, the law can be verified. At any point of time, the angular momentum can be given as $L = mr^2\omega$.

Now, consider a small area ΔA described in a short time interval Δt and the covered angle is $\Delta\theta$. Let the radius of curvature of the path be r, then the length of the arc covered = $r\,\Delta\theta$

$\Delta A = 1/2[r.(r.\Delta\theta)] = 1/2r^2\Delta\theta$

Therefore, $\Delta A/\Delta t = [\,1/2r^2]\Delta\theta/dt$

Taking limits on both sides as, $\Delta t \to 0$, we get,

$$\lim_{\Delta t\to 0}\frac{\Delta A}{\Delta t} = \lim_{\Delta t\to 0}\frac{1}{2}r^2\frac{\Delta\theta}{\Delta t}$$
$$\frac{dA}{dt} = \frac{1}{2}r^2\omega$$
$$\frac{dA}{dt} = \frac{L}{2m}$$

Now, by conservation of angular momentum, L is a constant.

Thus, dA/dt = Constant

- **Kepler's Third Law – The Law of Periods**
- According to Kepler's law of periods,"The square of the time period of revolution of a planet around the sun in an elliptical orbit is directly proportional to the cube of its semi-major axis".

$$\mathbf{T^2 \propto a^3}$$

- The shorter the orbit of the planet around the sun, the shorter the time taken to complete one revolution. Using the equations of Newton's law of gravitation and laws of motion, Kepler's third law takes a more general form.
- $P^2 = 4\pi^2 / [G(M_1 + M_2)] \times a^3$
- Where M_1 and M_2 are the masses of the two orbiting objects in solar masses.